Er.Roshani Giri

IT Engineer | Backend Specialist

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PROFESSIONAL SUMMARY

Backend Developer with professional experience building and maintaining production-ready applications using Node.js, Express.js, RESTful APIs, and relational databases. Experienced in developing secure backend systems, implementing business logic, integrating third-party services, and collaborating with cross-functional teams on live projects. Additionally, completed three months of Data Science training and worked on machine learning, data analysis, and computer vision projects using Python, Pandas, NumPy, scikit-learn, and OpenCV. Interested in combining software engineering, data-driven problem-solving, and AI/ML to build practical, scalable applications.

WORK EXPERIENCE

Backend Developer |*Blackmoon Technology | Lalitpur, Nepal* | *May 2025 – Present*

- Built highly scalable backend systems and implemented RESTful APIs with Node.js and Express.js in real-time enterprise-level applications.
- Led the backend development of the Agnikunda application by incorporating business logic, security mechanisms such as authentication and authorization, as well as effective data flows.
- Developed an application module for a Hospital Management System including patient record, doctor management, appointment, and data processing features.
- Developed and optimized the database schema (both SQL and NoSQL databases) and ensured high-performance database queries.
- Ensured implementation of best practices regarding validation and error handling.
- Developed third-party integrations as well as efficient file management solutions such as storing files via Cloudinary/Firebase services.
- Worked closely with frontend development teams to develop efficient APIs and ensure proper system integration.

Data Science

- Practiced data cleaning, preprocessing, exploratory data analysis (EDA), and visualization using Python.
- Worked with Pandas, NumPy, Matplotlib, and Seaborn to analyze datasets and identify meaningful patterns.
- Explored supervised machine learning techniques for classification and regression problems using scikit-learn.
- Worked on machine learning and computer vision projects involving dataset preparation, model development, and result evaluation.
- Strengthened practical knowledge of statistical analysis, feature engineering, and data-driven problem-solving.

EDUCATION

Bachelor of Engineering in Information Technology(BE.IT)

Pokhara University ,Everest Engineering College, Lalitpur, Nepal | 2022-*2026*

TECHNICAL SKILLS

Backend Development: Node.js, Express.js, RESTful API Design, Python, Firebase Functions

Frontend Development: React.js, JavaScript (ES6+), HTML5, CSS3, Responsive Design

AI/ML & Data Science: Machine Learning, Data Analysis, Data Cleaning, Exploratory Data Analysis (EDA), scikitlearn, Pandas, NumPy, Matplotlib, Seaborn

Computer Vision: OpenCV, SIFT, DBSCAN, Image Processing, Image Forgery Detection

Frontend Development: React.js, JavaScript (ES6+), HTML5, CSS3, Responsive Design

Databases: MySQL, Firebase Realtime Database

Cloud & Storage: Cloudinary (Image Storage), Firebase

Design & Prototyping: Figma (UI Review, Design Handoff, Component Reference), Canva

Tools & Platforms: Git, GitHub, Postman, VS Code, Jira, Jupyter Notebook

Soft Skills: Problem Solving, Team Collaboration, Agile/Scrum, Cross-team Communication

PROJECTS

Copy-Move Image Forgery Detection (CMFD) | CMFD

Academic Project | Technologies: Python, OpenCV , SIFT, DBSCAN (scikit-learn), Numpy

- Built a SIFT-based key-point detection system using OpenCV to identify duplicated regions in images.
- Used DBSCAN clustering to group similar key-points and detect copy-move forgeries.
- Generated forgery masks and annotated images to visually highlight tampered areas.
- System can also classify images as clean when no valid forgery clusters are found.

Deadstock- MERN Stack E-Commerce Web App

Academic Projects | Technologies: Express.js, React.js, Node.js, Firebase Auth, Cloudinary

- Developed full-stack e-commerce platform with product listing, cart management, and order workflow using the MERN stack.
- Implemented Firebase Authentication for secure user login and session management, alongside Cloudinary for scalable product image storage and delivery.
- Built RESTful APIs for product CRUD operations, user management, and order processing.

Agnikunda — Online Examination Platform

Client Project | Technologies: Node.js, Express.js, MySQL, RESTful APIs, JWT, Nodemailer

- Developed backend RESTful APIs for an online examination platform supporting student registration, authentication, and instructor management.
- Implemented category and sub-category management, question set organization, and support for different question types.
- Built subscription-based content access with time-based expiry, purchase history, and repurchase validation.
- Implemented quiz score tracking, performance history, and OTP-based password reset functionality.

Sahaj Yatri — Travel Management Platform

Project | Technologies: Node.js, Express.js, MySQL, RESTful APIs

- Contributed to backend development for a travel management platform designed to support travel-related services and information management.
- Developed and integrated RESTful APIs for data handling and communication between the client and server.
- Worked with database operations and backend business logic to support application functionality.

Hospital Management System (HMS)

Professional Project | Technologies: Node.js, Express.js, JavaScript, MySQL, RESTful APIs

- Developed and maintained backend RESTful APIs for a hospital management system supporting patient registration, OPD/IPD visits, consultations, and admissions.
- Implemented business logic for billing, invoices, pharmacy requests, laboratory services, and patient medical records.
- Integrated database operations and validation to maintain data consistency across hospital workflows.
- Debugged and resolved Jira-reported issues, improving API reliability and supporting ongoing system maintenance.

LANGUAGES

English (Professional Proficiency) **Nepali** (Native) **Hindi** (Conversational)